

Student Performance Data Explorer

AI/ML Internship — Task 1 | Exploratory Data Analysis Report

This report documents the EDA performed on the provided student performance dataset. The analysis covers data inspection, cleaning, descriptive statistics and visual exploration.

Dataset: 20 students, 7 columns
Missing values after cleaning: 0
Duplicate rows after cleaning: 0

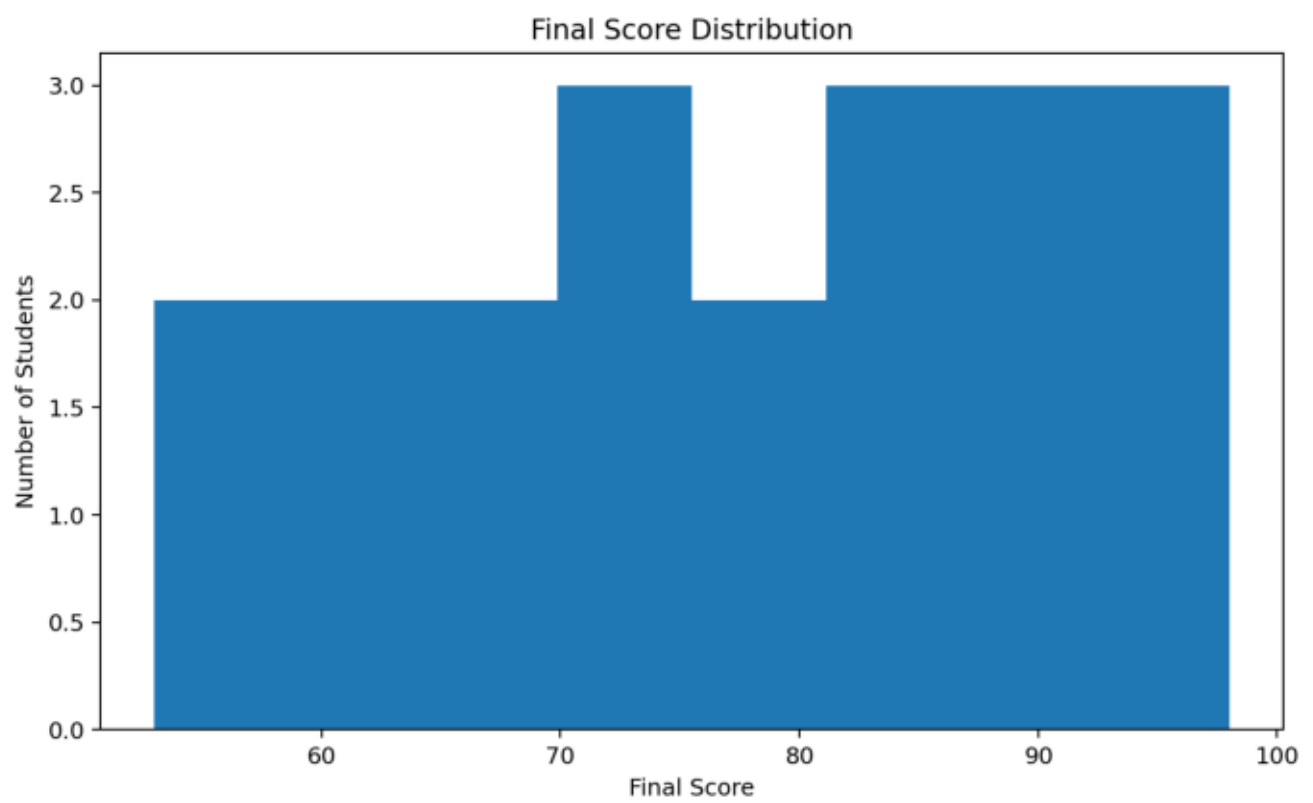
Final score – mean: 77.55, median: 77.50, mode: 53.00
Average study hours: 4.42
Average attendance: 84.65%

Correlations with final score:
Study hours: 0.99
Attendance: 0.98
Previous score: 1.00
Assignments completed: 0.97

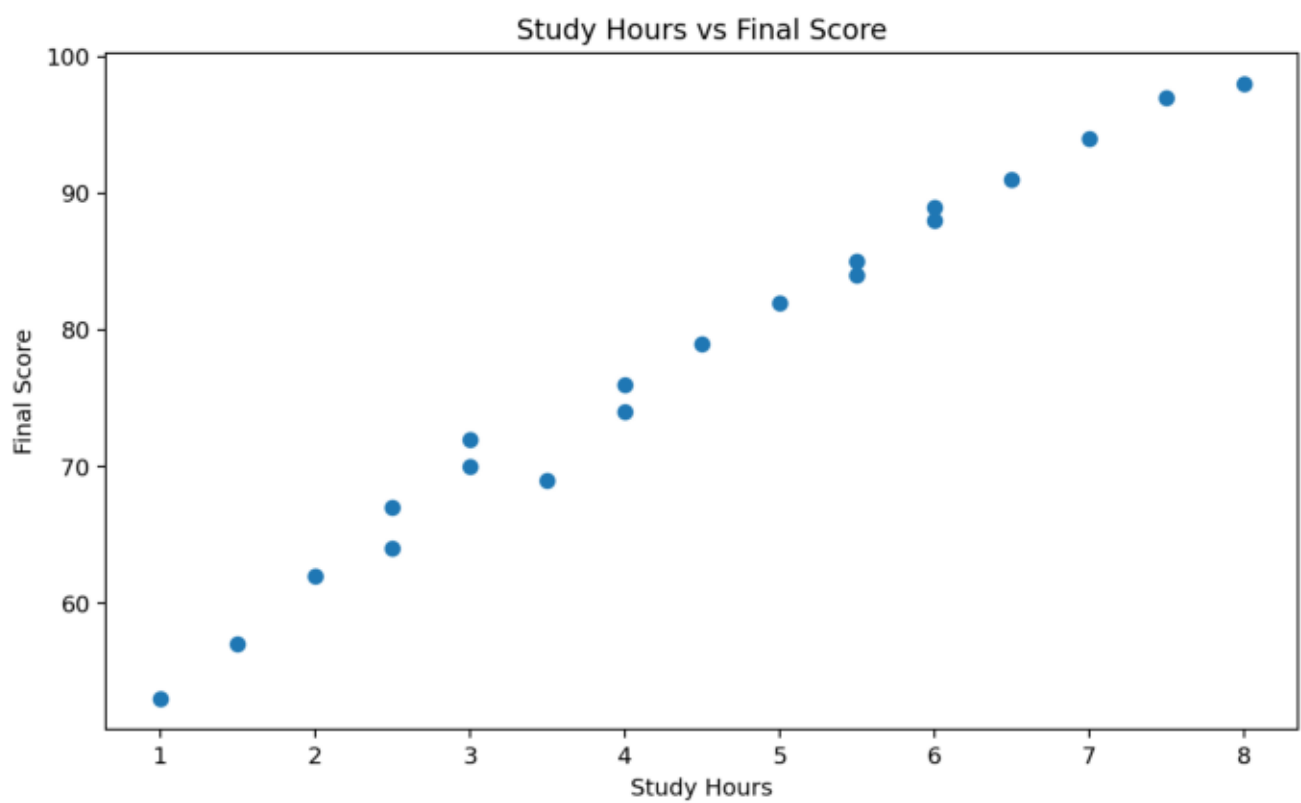
Key findings

- Final scores average 77.55, with a median of 77.50.
- Study hours have a correlation of 0.99 with final score in this dataset.
- Attendance has a correlation of 0.98 with final score.
- Previous score has a correlation of 1.00 with final score.
- Assignments completed have a correlation of 0.97 with final score.
- Correlation describes association in this dataset; it does not establish causation.

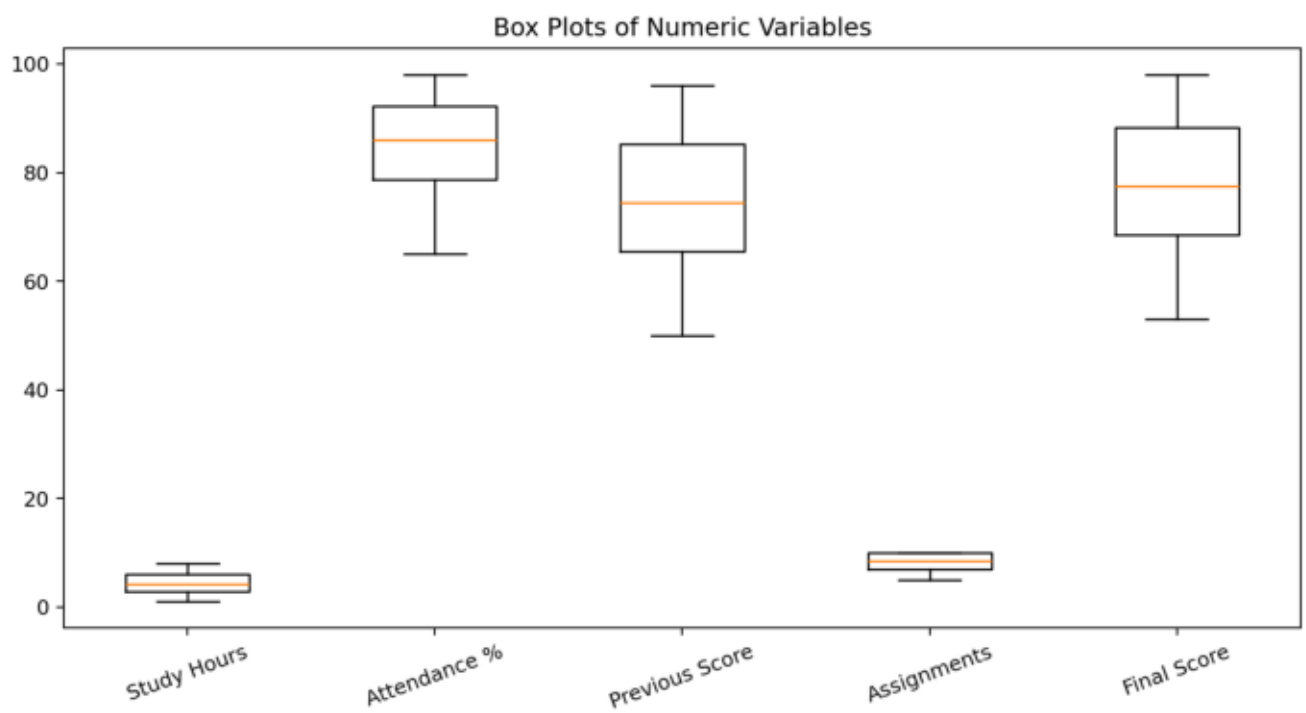
Final Score Distribution



Study Hours vs Final Score



Numeric Variables — Box Plots



Correlation Heatmap

